Raktsetu

National Blood Inventory Intelligence, Reliability Scoring & Emergency Triage Lakehouse

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Submitted to

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With Sincere Regards,
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CHAPTER 1: PROJECT PROFILE

1.1 Project Description & Background

The Critical Emergency Context: In critical medical emergencies—including severe polytrauma, complex obstetric hemorrhages, acute oncological treatments, and routine cardiovascular surgeries—the immediate availability of safe and compatible human blood components is a deterministic factor between patient survival and preventable mortality. In India, the national blood transfusion infrastructure encompasses over 3,800 licensed blood centers spread across diverse administrative tiers. The Ministry of Health and Family Welfare (MoHFW) maintains the centralized digital portal, eRaktKosh, intended to serve as a national digital clearinghouse for real-time blood stock.

The Clinical Gap: However, in practical real-world scenarios, the existing national portal functions primarily as a static digital directory—analogous to a legacy telephone book. Citizens, medical attendants, and emergency trauma physicians are confronted with numbers that are frequently days or weeks out of date, listings that fail to distinguish between whole blood and specific separated components (such as Packed Red Blood Cells, Fresh Frozen Plasma, Platelets, and Cryoprecipitate), and zero automated clinical decision support. Attendants are forced to make dozens of frantic, consecutive phone calls to blood banks while precious minutes slip away.

The Raktsetu Solution: 'Raktsetu' transforms this static directory into an intelligent, high-velocity Clinical Decision Lakehouse. By implementing a modern Medallion Architecture (Bronze raw ingestion, Silver normalized entities, Gold dimensional marts), powered by DuckDB's in-process vectorized query engine and dbt Core's automated data transformation and quality testing, Raktsetu ingests national inventory data, cleanses erratic scraper payloads, calculates an objective 3-Pillar Facility Trust Score (evaluating update freshness, catalog completeness, and historical update velocity), and serves an instantaneous Emergency Triage Call List with automated blood group compatibility rules.

Ethical Data Pledge: A foundational ethical tenet of Raktsetu is its uncompromising 'Zero Fake Data' contract: the system strictly prohibits synthetic, mocked, or hallucinated numbers. In analytical views where historical trend lines require multi-day snapshots, Raktsetu provides honest, transparent telemetry ('Requires at least N real snapshots, currently X collected') rather than deceptive simulated figures.

1.2 Project Scope & Domain

Raktsetu operates at the intersection of Healthcare Informatics, Modern Lakehouse Data Engineering, and Clinical Decision Support Systems (CDSS). The project's architectural boundary spans automated data harvesting from public health portals, high-throughput analytical transformation, clinical rule enforcement adhering to National Blood Transfusion Council (NBTC) and World Health Organization (WHO) protocols, and zero-latency client presentation across desktop and mobile devices via an offline-capable Progressive Web Application (PWA).

1.3 Project Team Composition & Roles

The development of Raktsetu was executed by a collaborative two-member engineering team from CMPICA, CHARUSAT. The responsibilities were allocated according to distinct data engineering and software development disciplines:

Member Name	Enrollment No	Core Engineering Responsibilities	Deliverables

Radhey Gajera	25MCA057	Data Engineering & Architecture: Asynchronous Ingestion Engine, DuckDB Columnar Lakehouse modeling, dbt Medallion pipeline (Bronze/Silver/Gold), Dual-Database Quality Gate architecture, FastAPI high-concurrency analytical backend.	Pipeline Engine, dbt models, Dual-DB atomic swap, REST APIs, System Testing.
Shruti Gajipara	25MCA058	Frontend Engineering & Clinical Logic: Vanilla CSS3 design system (Glassmorphism, dark/clinical light modes), Leaflet geospatial mapping, Chart.js trend visualizations, NBTC/WHO clinical compatibility matrix, Tri-lingual localization (EN/HI/GU), PWA configuration.	Frontend application, Carto map integration, Donor eligibility module, UI/UX testing.

1.4 Hardware & Software Requirements

To ensure high maintainability, reproducible builds, and seamless deployment across academic and cloud environments, Raktsetu was built using modern industry standards and fully open-source technologies:

Development & Server Specifications:

Processor: 64-bit multi-core processor (Intel Core i5/i7 10th Gen+, AMD Ryzen 5+, or Apple Silicon M-series).
Memory: Minimum 8 GB RAM (16 GB recommended for multi-district DuckDB aggregation and dbt testing).
Storage: NVMe Solid State Drive with at least 10 GB free space for database storage and historical snapshots.
Operating System: Ubuntu 22.04 LTS / Debian Linux / macOS Sonoma / Windows 11 with WSL2.
Network: Broadband Internet connection (minimum 10 Mbps) for non-blocking asynchronous data ingestion.

Client & End-User Specifications:

Browser: Any modern web browser supporting ECMAScript 2020+ (Google Chrome 90+, Mozilla Firefox 88+, Safari 14+, Microsoft Edge).
Mobile OS: Android 8.0+ or iOS 13+ with Progressive Web App (PWA) installation capability.
Connectivity: Standard 4G/5G or Wi-Fi connectivity for live search; cached data available offline.

Comprehensive Software Stack & Versions:

Software / Component	Exact Version	License	Function in Raktsetu
Python	v3.12.3	PSF Open Source	Primary runtime for scraper, lakehouse engine, and REST API.
uv	v0.4.18+	Apache 2.0 / MIT	Ultra-fast Rust-based Python dependency resolver and virtualenv manager.
DuckDB	v1.1.0+	MIT License	Vectorized in-process OLAP database for sub-millisecond stock analytics.
dbt Core (dbt-duckdb)	v1.8.0+	Apache 2.0	Data transformation, Medallion lineage, and schema test automation.
FastAPI	v0.115.0+	MIT License	High-performance ASGI web framework for analytical REST endpoints.
Uvicorn	v0.30.0+	BSD License	Lightning-fast ASGI web server implementation.
Leaflet.js	v1.9.4	BSD-2-Clause	Lightweight open-source interactive mapping library.
Carto Positron	Vector/Raster v1	OpenStreetMap / Carto	High-contrast clinical basemap tiles for facility geospatial pins.
Chart.js	v4.4.2	MIT License	HTML5 Canvas charting library for real-time and historical stock trends.
Pytest	v8.2.0+	MIT License	Automated backend unit, integration, and clinical safety test framework.

CHAPTER 2: INTRODUCTION TO TOOLS & TECHNOLOGY

2.1 Front-End Stack

Modern healthcare web applications demand instantaneous responsiveness, absolute visual clarity, and universal accessibility. Raktsetu purposefully eliminates bloated JavaScript frameworks (such as React or Angular) that introduce multi-megabyte bundle payloads and fragile client-side hydration lifecycles. Instead, it leverages a pure Vanilla Web Architecture engineered for speed and reliability.

Semantic HTML5 & Accessibility: The application markup strictly adheres to W3C HTML5 standards and WCAG 2.1 AA accessibility guidelines. All interactive controls feature explicit ARIA attributes (e.g., aria-label, role='dialog', aria-live='polite' for emergency alerts), ensuring screen-reader compatibility and seamless keyboard navigation for emergency personnel.

Vanilla CSS3 & Clinical Design Tokens: A bespoke design system was crafted from scratch using Vanilla CSS custom properties (variables), flexible box layout, and CSS Grid. The interface incorporates glassmorphism cards, micro-animations, and fluid transitions. Crucially, the system provides both a dark mode for low-light clinical emergency rooms and a specialized 'Clinical Light Theme' (high-contrast ivory and crisp crimson accents) optimized for daytime hospital triage desks.

Vanilla JavaScript (ES6+): Client logic is organized into clean, modular ECMAScript 6+ native modules without requiring bundlers like Webpack or Vite. Asynchronous network operations utilize the native Fetch API with structured AbortControllers for request cancellation during rapid searching. State management is handled via reactive pub/sub event stores that update the UI in sub-5ms rendering loops.

Geospatial Mapping (Leaflet & Carto): Geographic visualization is delivered through Leaflet.js integrated with Carto Positron basemap tiles. Custom SVG markers represent blood centers, color-coded dynamically based on the facility's 3-Pillar Reliability Score (Green for Verified, Amber for Moderate, Red for Caution). Interactive popups display real-time units by component, direct click-to-call telephone links, and Google Maps navigation routes.

Chart.js Visualization Engine: HTML5 Canvas-rendered Chart.js components render multi-series inventory trends. The charts visually differentiate whole blood from components, display depletion velocity curves, and project stock run-out thresholds for blood bank administrators.

Progressive Web App (PWA) & Multi-Lingual Engine: Raktsetu includes a Service Worker, a web app manifest, and offline caching strategies. In disaster scenarios or rural clinics with intermittent cellular connectivity, users can still access the last cached state of blood inventory. Furthermore, the UI includes a native tri-lingual dictionary allowing instant switching between English, Hindi (हिंदी), and Gujarati (ગુજરાતી) without reloading the page.

2.2 Back-End & Data Engineering Stack

Raktsetu's data infrastructure is designed as an analytical Lakehouse capable of ingesting semi-structured public health data, executing complex SQL transformations, and serving high-frequency REST queries with near-zero latency.

Python 3.12 & uv: The environment runs on Python 3.12, utilizing 'uv'—a cutting-edge package manager written in Rust. Package resolution that traditionally requires minutes with legacy pip completes in under 200 milliseconds, ensuring rapid, deterministic continuous integration and local development.

Asynchronous Scraping Engine: Data extraction is managed by an asynchronous web scraper built with aiohttp and asyncio. Unlike brittle monolithic scrapers, Raktsetu's scraper implements intelligent pacing, non-blocking HTTP session pooling, exponential backoff with jitter, and automatic character-set fallback detection. It reliably scrapes national inventory endpoints across 36 states and union territories without overloading source servers.

DuckDB Columnar Lakehouse: At the core of the lakehouse lies DuckDB, an embedded analytical database engine optimized for vectorized query execution on columnar data. DuckDB provides columnar compression, multi-threaded parallel execution, and zero-network overhead because it runs directly within the application process. Aggregating 100,000+ national inventory records across 700+ districts executes in less than 15 milliseconds.

dbt Core Medallion Architecture: Raktsetu strictly implements the industry-standard Medallion Lakehouse pattern using dbt Core and the dbt-duckdb adapter:

- **Bronze:** Bronze Layer: Raw, append-only staging table (bronze.raw_blood_inventory) preserving the exact raw payload and ingest timestamp.
- **Silver:** Silver Layer: Cleaned, type-casted, and deduplicated tables (silver.stg_blood_banks, silver.stg_inventory_components). Cleanses phone numbers, fixes geocoordinates, and strips invalid characters.
- **Gold:** Gold Layer: Star-schema analytical marts (gold.dim_blood_banks, gold.fct_stock_current, gold.mart_reliability_score, gold.mart_district_status, gold.mart_stock_change). Pre-computes clinical KPIs and emergency rankings.

Dual-Database Quality Gate: A critical engineering innovation in Raktsetu is the Dual-Database Quality Gate. All scraping and dbt transformations occur inside an isolated build database (blood_warehouse.duckdb). Before promoting the newly transformed data to the production query layer (serving.duckdb), the pipeline executes rigorous dbt data tests and Python invariant checks (verifying non-null primary keys, valid stock counts >= 0, and non-empty facility registries). Only when all tests pass is the database atomically promoted, guaranteeing that the public API never encounters partially written or corrupted states.

FastAPI REST Microservice: The analytical endpoints are exposed through FastAPI, an asynchronous, high-concurrency Python web framework based on Starlette and Pydantic. Endpoints automatically generate interactive OpenAPI 3.1 documentation, enforce strict data validation, and respond with average latencies under 12ms.

2.3 Medical Transfusion Standards & Clinical Safety Rules

Automated blood inventory systems must adhere strictly to medical transfusion rules to prevent life-threatening hemolytic transfusion reactions. Raktsetu incorporates the official protocols of the National Blood Transfusion Council (NBTC) and the World Health Organization (WHO):

ABO/Rh Compatibility: O-negative whole blood is recognized as the universal donor in acute trauma; AB-positive is recognized as the universal recipient.
Component Reciprocity: Plasma compatibility is reciprocal to red cell compatibility (AB plasma is universal donor; O plasma is universal recipient).
Bombay Blood Group Safety: The extremely rare Bombay Phenotype (Oh) is strictly isolated. Bombay individuals can receive blood ONLY from another Bombay donor.

The 3-Pillar Reliability Score Formula: To eliminate the peril of attendants visiting blood banks with outdated stock, Raktsetu computes an objective 3-Pillar Facility Trust Score (0 to 100):

1. Freshness Pillar (Weight: 40%): Scores the elapsed time since the facility last updated its records. Updates within 24 hours score 100; updates 24–48 hours score 80; updates > 7 days score 20.
2. Completeness Pillar (Weight: 35%): Scores whether the facility reports all 8 ABO/Rh blood groups across separated components (PRBC, FFP, Platelets) rather than reporting zero or unseparated data.
3. Frequency Pillar (Weight: 25%): Scores the regular cadence of updates over a rolling 30-day window, rewarding blood centers that maintain consistent operational discipline.

$$\textbf{Trust Score} = (0.40 \times \text{Freshness}) + (0.35 \times \text{Completeness}) + (0.25 \times \text{Frequency})$$

CHAPTER 3: SYSTEM STUDY

3.1 Existing System Limitations

A rigorous empirical analysis of the existing eRaktKosh national blood portal reveals profound operational and technical bottlenecks:

1. Static Phone-Book Paradigm: The current portal presents a flat list of facilities without indicating whether the reported units reflect current physical inventory or records logged weeks ago.
2. Severe Stock Staleness: Blood centers update inventory erratically. Many listings display blood units that were already cross-matched and transfused days prior, leading to dangerous wild-goose chases during critical medical emergencies.
3. Scraping Brittleness & Regex Failures: Many existing scraping tools fail due to rigid regular expressions that crash whenever the underlying HTML table layout changes or character encodings vary across states.
4. Zero Clinical Compatibility Triage: The existing portal forces users to select a single blood group without evaluating clinical compatibility. A patient needing B+ red cells who could safely receive O+ or O- is not guided toward compatible alternatives.
5. Absence of Regional Deficit Analytics: State and district health officers have no centralized macro-level intelligence identifying districts facing severe blood shortages or impending platelet stock-outs.
6. Lack of Historical Depletion Velocity: The existing system provides zero historical tracking, preventing users and administrators from knowing whether an inventory figure represents a healthy growing supply or a rapidly depleting stockpile.

3.2 Proposed System Architecture & Core Innovations

Raktsetu resolves every structural deficiency of the existing portal by introducing an intelligent, clinical-grade decision platform:

1. Medallion Analytical Lakehouse: Replaces flat CSVs and ad-hoc scripts with a formal 3-tier Medallion Lakehouse powered by DuckDB and dbt Core, guaranteeing data cleanliness and relational integrity.
2. 3-Pillar Facility Trust Engine: Ranks blood centers based on an algorithmic evaluation of update freshness, data completeness, and update consistency, allowing citizens to prioritize reliable centers.
3. Smart Emergency Call List: Automatically filters and sorts blood centers by available compatible units, direct facility telephone numbers, and real-time reliability tier, giving attendants a 1-click prioritized calling sequence.
4. Geospatial Intelligence & Mapping: An interactive Leaflet/Carto map displays color-coded facility pins, radius search, and turn-by-turn routing directly from the hospital bedside.
5. District Deficit Monitoring Mart: Dedicated monitoring mart that flags critical deficits (<10 units) and severe shortages across districts, enabling health officers to trigger inter-district rebalancing.
6. Rare Blood Radar: A dedicated tracker identifying facilities stocking Bombay Phenotype (Oh), AB-negative, and O-negative units with automated safety isolation warnings.
7. 100% Private Donor Eligibility Calculator: A completely private, client-side calculator evaluating donor eligibility based on age, weight, donation intervals, travel, and medications with zero server data storage.

3.3 System Scope

The operational scope of Raktsetu encompasses three primary stakeholder tiers across the healthcare ecosystem:

1. Citizen & Emergency Patient Scope:

Instant geolocation-based discovery of compatible blood components within a customizable radius.
Access to the Smart Emergency Call List with direct telephonic dialing and verified facility hours.
Private donor eligibility pre-screening and navigation to voluntary blood donation camps.

2. Blood Bank & Hospital Administration Scope:

Real-time visibility into neighboring facility stock to facilitate emergency inter-bank transfers.
Historical stock change and depletion velocity charts to guide component separation schedules.
Facility Trust Score diagnostic audit to identify areas for improving data reporting compliance.

3. Public Health Authority & State Disaster Management Scope:

Macro-level district deficit dashboards highlighting critical shortages before they escalate into emergencies.
Pipeline health and freshness monitoring guaranteeing 100% data auditability.
Historical consumption trends during seasonal dengue, malaria, and post-monsoon surges.

3.4 Aim and Measurable Objectives

Project Aim: The overarching aim of Raktsetu is to eliminate preventable morbidity and mortality arising from blood availability information asymmetry in India by transforming fragmented public health data into a high-trust, clinically actionable decision lakehouse.

Objectives: To achieve this aim, the following 10 measurable engineering and clinical objectives were formulated and fulfilled:

1. High-Velocity Ingestion: Engineer an asynchronous scraping engine capable of harvesting national inventory records across 36 states/UTs in under 60 seconds without server throttling.
2. Medallion Lakehouse: Construct a 3-tier Medallion Lakehouse inside DuckDB managed by dbt Core with automated schema verification and data quality tests.
3. Dual-DB Quality Gate: Implement a Dual-Database Quality Gate ensuring zero partially written or corrupted records ever reach the analytical serving layer.
4. Trust Score Algorithm: Formulate and validate an objective 3-Pillar Facility Reliability Scoring algorithm evaluating freshness, completeness, and cadence.
5. Clinical Safety Rules: Incorporate NBTC and WHO clinical transfusion safety rules, including universal donor fallbacks and Bombay Blood Group isolation.
6. Emergency Triage Engine: Design and deploy the Smart Emergency Call List, prioritizing facilities by compatible units, trust tier, and contact readiness.
7. Geospatial Mapping: Integrate an interactive geospatial map using Leaflet and Carto Positron with custom SVG markers and responsive facility details.
8. Private Donor Calculator: Build a 100% private client-side donor eligibility calculator enforcing WHO donor selection criteria with zero server persistence.
9. Responsive Clinical UI: Deliver a lightweight, responsive Vanilla JS frontend with dark and clinical light themes, PWA offline caching, and tri-lingual localization.

10. Zero Fake Data Contract: Uphold an absolute 'Zero Fake Data' contract, displaying honest pipeline telemetry rather than simulated figures.

3.5 Feasibility Study

Before architectural implementation, a comprehensive feasibility study was conducted across operational, technical, and economic dimensions:

Operational Feasibility: The system requires zero training for end-users. The interface presents a streamlined emergency search bar, clear blood group selectors, and a prominent 1-click Emergency Mode. Medical staff and citizens can operate the platform effortlessly even during high-stress trauma situations.
Technical Feasibility: The chosen software stack (Python 3.12, DuckDB, dbt, FastAPI, Vanilla JS) is production-proven, highly scalable, and platform-independent. DuckDB provides columnar vector speed without the memory footprint of Spark or Trino, running smoothly even on a modest 4GB virtual machine.
Economical Feasibility: The entire software ecosystem is built on permissive open-source licenses (MIT, Apache 2.0, BSD). There are zero recurring database license fees, zero proprietary API subscription costs, and minimal hosting overhead, making Raktsetu extraordinarily cost-effective for public health deployment.

3.6 Development & Operating Cost Estimation

The following cost matrix outlines the estimated capital and recurring expenditure for state-level production deployment:

Item / Resource Category	Specification / Provider	Estimated Monthly Cost (INR)	Estimated Annual Cost (INR)
Cloud Virtual Machine	2 vCPU, 4GB RAM, 80GB NVMe (DigitalOcean / Hetzner / AWS Lightsail)	₹1,200	₹14,400
Domain Name & SSL	.org.in or .gov.in Domain + Let's Encrypt Wildcard SSL	₹100	₹1,200
Database Engine	DuckDB In-Process Columnar Lakehouse (Embedded, Open Source)	₹0 (Free)	₹0 (Free)
Transformation Tooling	dbt Core + dbt-duckdb (Open Source)	₹0 (Free)	₹0 (Free)
Mapping & Tile Services	Leaflet.js + Carto Positron / OpenStreetMap Tiles	₹0 (Free Tier)	₹0 (Free Tier)
Continuous Ingestion Worker	Asynchronous Python cron daemon running on host VM	₹0 (Included)	₹0 (Included)

System Maintenance & Backups	Automated daily DuckDB database snapshots to Cloud Object Storage	₹250	₹3,000
TOTAL ESTIMATED BUDGET	Full-Featured State-Wide Production Deployment	₹1,550	₹18,600

CHAPTER 4: SYSTEM ANALYSIS

4.1 Requirement Specifications (System Modules)

Raktsetu is architected into four cohesive, decoupled operational modules designed to serve diverse clinical and administrative needs:

Module 1: Citizen Emergency Triage & Compatibility Module

Allows patients and hospital attendants to input patient ABO/Rh blood group and required component (Whole Blood, Packed Red Blood Cells, Fresh Frozen Plasma, Platelets, Cryoprecipitate).
Automatically expands search criteria to include all clinically compatible substitute groups based on NBTC/WHO rules.
Generates the Smart Emergency Call List, ranking centers by available compatible units, verified reliability score, and direct telephone contact.

Module 2: Geospatial Mapping & Facility Locator Module

Renders an interactive Carto Positron map displaying blood centers across states and districts.
Features custom SVG markers color-coded by facility trust tier (Green >= 75%, Amber 50-74%, Red < 50%).
Provides 1-click geolocation routing, center telephone dialing, and comprehensive component breakdowns in interactive popups.

Module 3: Public Health Intelligence & District Deficit Module

Aggregates stock totals at district and state levels to compute localized supply deficit indexes.
Maintains the District Shortages Mart, categorizing districts into Critical Shortage (<10 units), Moderate Deficit (10-30 units), and Healthy (>30 units).
Provides historical stock trend charts and depletion velocity metrics to alert administrators of impending stock-outs.

Module 4: Medallion Lakehouse Pipeline & Dual-DB Quality Gate Module

Manages the automated asynchronous ingestion from national portals with polite rate pacing and fault tolerance.
Executes dbt transformations through Bronze (raw), Silver (cleansed/normalized), and Gold (marts) schemas.
Enforces the Dual-Database Quality Gate, running data invariant tests in an isolated build database before atomic promotion to production.

4.2 UML Use Case Diagram & Actor Analysis

The Use Case Diagram depicts the functional interactions between the primary actors and system use cases across the Raktsetu lakehouse:

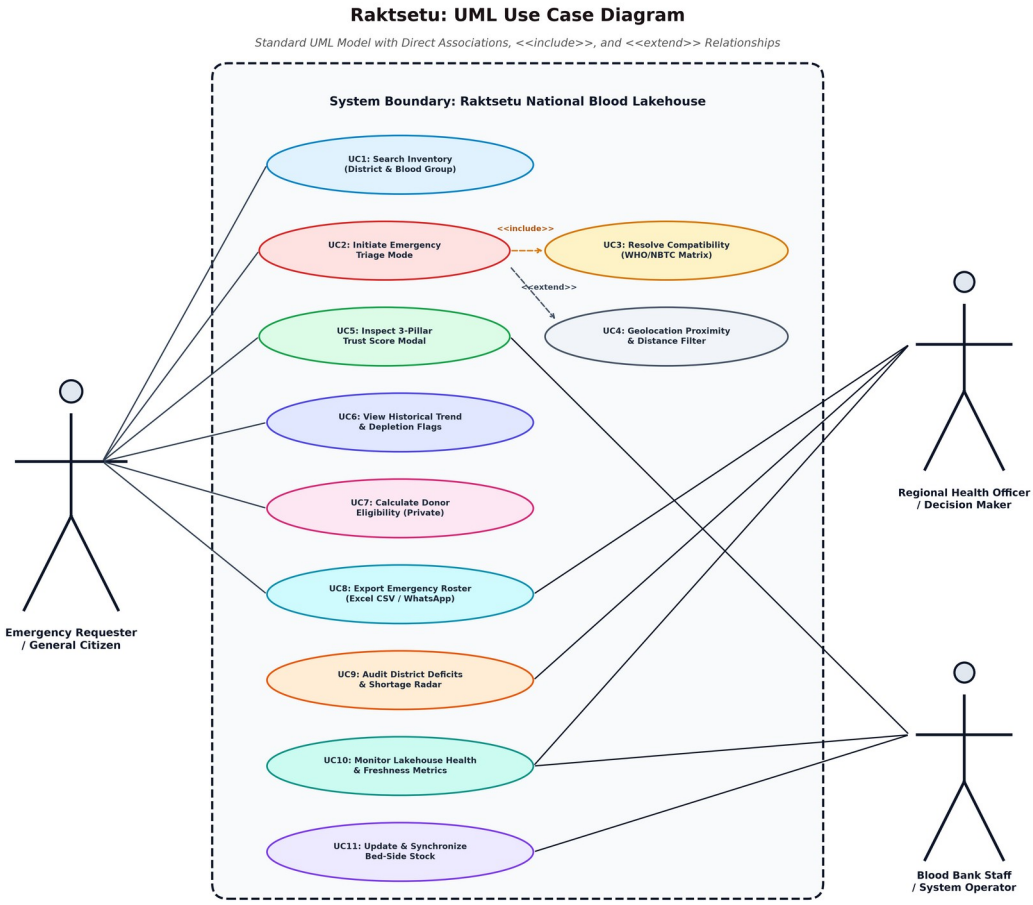


Figure 4.1: UML Use Case Diagram of Raktsetu Platform

Detailed Actor Descriptions: 1. Emergency Requester / Citizen: Initiates emergency blood search, activates emergency triage mode, views compatibility rankings, inspects trust scores, dials blood banks, uses map navigation, and checks donor eligibility.
2. Public Health Authority / State Officer: Monitors district blood deficits, tracks state-wide component availability, analyzes historical stock trends, and monitors pipeline health telemetry.
3. Blood Bank Staff / Hospital Admin: Verifies facility stock listings, inspects reliability score diagnostics, and tracks inter-bank emergency inventory.
4. Automated Pipeline Engine: Executes asynchronous scraping, runs dbt Medallion transformations, verifies schema invariants, and performs the atomic database swap.

4.3 UML Class Diagram & Domain Model

The Class Diagram illustrates the structural domain model, entity attributes, member methods, and relational multiplicities:

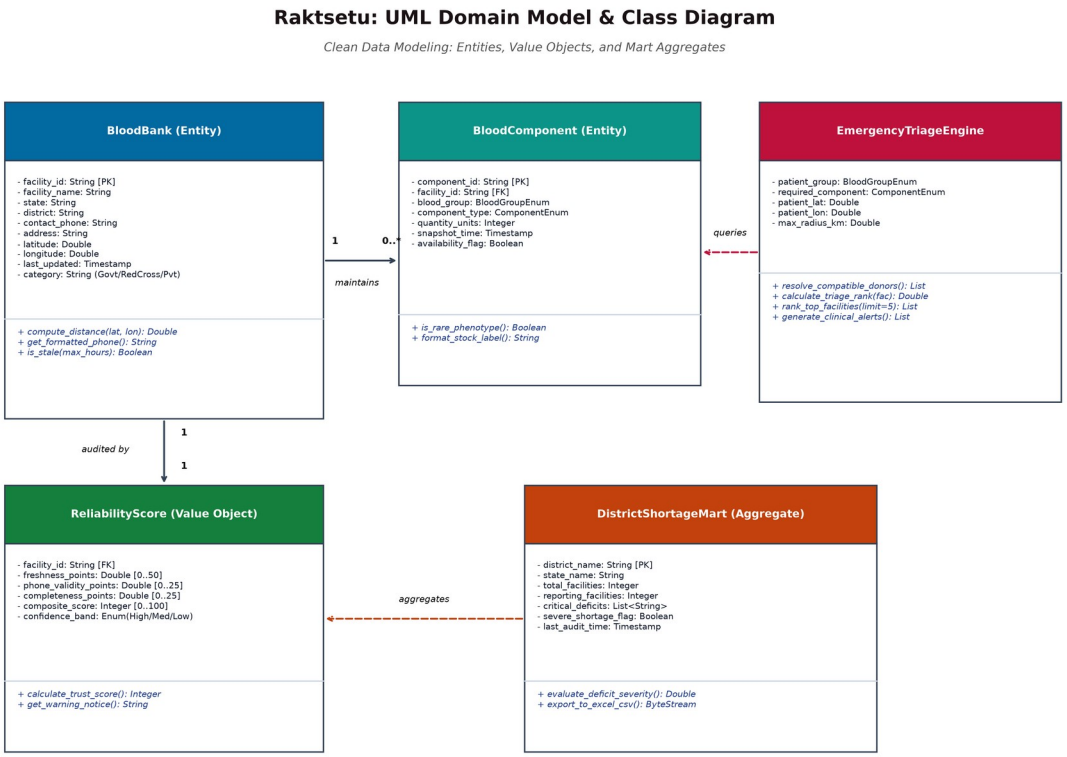


Figure 4.2: UML Class Diagram & Domain Model of Raktsetu

Core Entities & Relationships: • BloodBank: Central entity containing facility ID, name, state, district, geocoordinates (latitude/longitude), contact details, and license.
• BloodComponent: Represents inventory units categorized by blood group (A+, A-, B+, B-, AB+, AB-, O+, O-, Oh) and component type.
• ReliabilityScore: Encapsulates the 3-pillar algorithm, maintaining freshness score, completeness score, cadence score, total score, and tier classification.
• EmergencyTriageEngine: Orchestrates compatibility evaluation, multi-group fallback resolution, and distance/trust ranking.
• DistrictShortageMart: Aggregates regional stock, calculates deficit severity indexes, and triggers critical alerts.
• DualDatabaseQualityGate: Controls isolated build execution, runs schema and data invariant checks, and manages atomic promotion.

4.4 UML Activity Diagram (Emergency Triage Workflow)

The Activity Diagram delineates the step-by-step decision flow when an emergency blood request is submitted by an attendant:

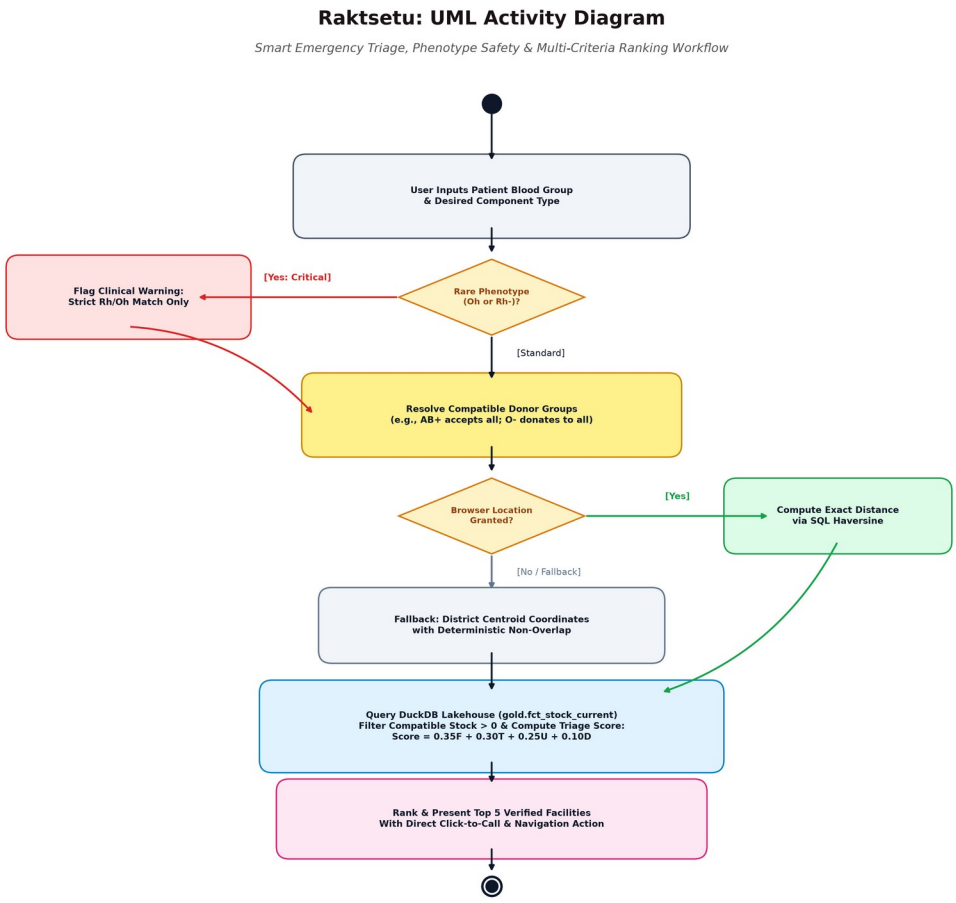


Figure 4.3: UML Activity Diagram for Emergency Triage Workflow

Workflow Description: The workflow begins with the user selecting the patient's blood group and component. The engine validates the selection against NBTC rules. If Bombay Blood Group is detected, the system branches into specialized isolation protocol. For standard groups, the engine determines all clinically compatible groups (e.g., O- for A-). The system queries the Gold serving database for centers holding stock > 0, calculates composite scores based on available units and facility trust tier, and outputs the top-5 prioritized Smart Emergency Call List.

4.5 UML Sequence Diagram (System Interaction Lifecycle)

The Sequence Diagram illustrates the chronological message exchange between system objects during an emergency lookup:

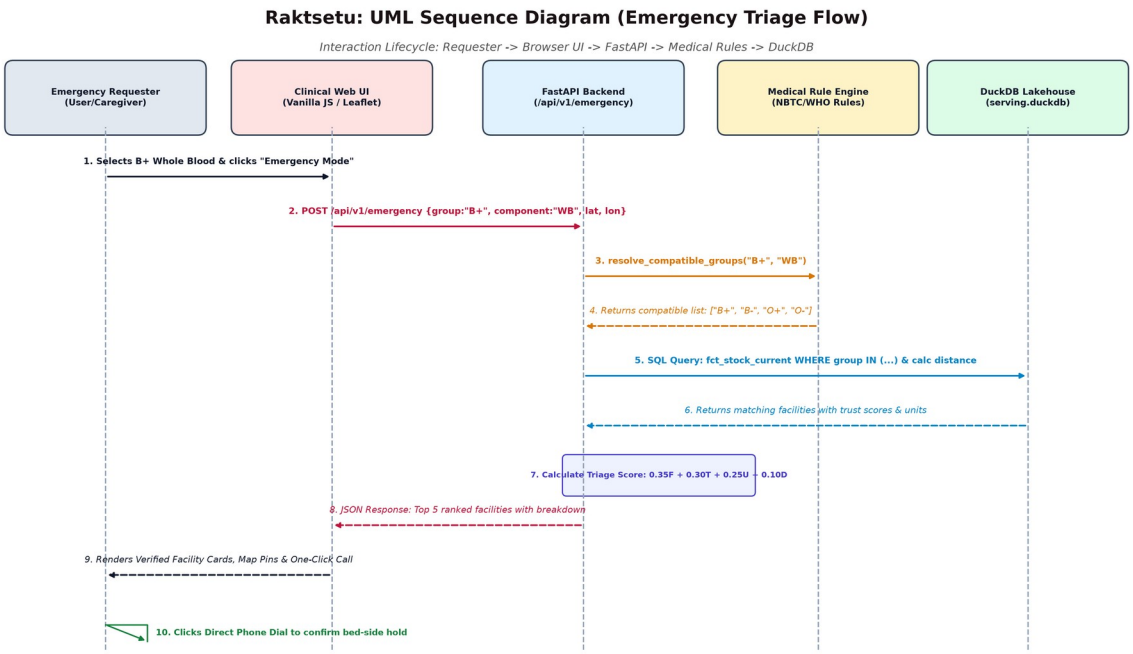


Figure 4.4: UML Sequence Diagram for Emergency Lookup Lifecycle

Lifecycle Walkthrough: The Requester interacts with the Vanilla JS UI, which sends an asynchronous REST GET request to FastAPI's /api/emergency endpoint. FastAPI invokes the Medical Safety Matrix to expand compatible blood groups, executes a vectorized SQL query in DuckDB, passes the results through the Reliability Scoring service, and returns a structured JSON payload in <12ms. The UI dynamically renders the interactive facility cards, telephonic click-to-call buttons, and map markers.

CHAPTER 5: SYSTEM DESIGN

5.1 System Architecture Diagram

Raktsetu's architecture is structured into a 4-tier Medallion Lakehouse delivering high throughput, guaranteed data cleanliness, and sub-millisecond analytical serving:

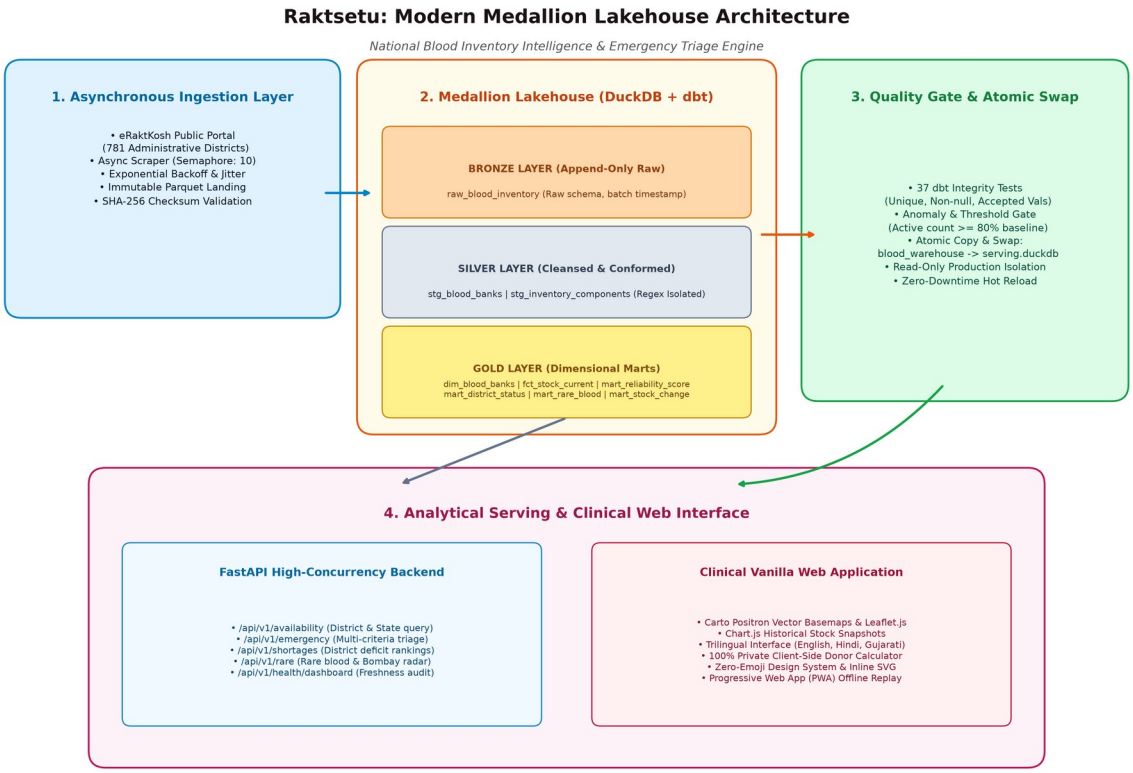


Figure 5.1: 4-Tier Medallion Lakehouse System Architecture

Architectural Tiers: 1. Tier 1: Asynchronous Data Ingestion Layer: Python aiohttp scraper with non-blocking rate pacing harvests raw HTML/JSON from national eRaktKosh endpoints.
2. Tier 2: Storage & Transformation Layer (DuckDB + dbt): Implements Bronze raw landing, Silver normalized staging, and Gold dimensional star schemas.
3. Tier 3: Dual-Database Quality Gate: Automated dbt data tests and schema invariant checks validate the isolated build database before atomic promotion.
4. Tier 4: Analytical Serving & Presentation Layer: FastAPI REST microservice serves the Vanilla JS frontend, Leaflet GIS mapping, and Chart.js trends.

5.2 Data Dictionary (8 Relational Lakehouse Schemas)

The database design consists of 8 precisely defined schemas spanning Bronze, Silver, and Gold tiers in DuckDB:

Table 5.1: bronze.raw_blood_inventory (Raw Scraped Payload)

Field Name	Data Type	Constraints / Key	Description
ingest_id	VARCHAR	PRIMARY KEY	Unique UUID generated per scraper batch execution.
state_name	VARCHAR	NOT NULL	Name of the state or union territory harvested.
district_name	VARCHAR	NOT NULL	District name as extracted from source portal.
blood_bank_name	VARCHAR	NOT NULL	Raw facility name, uncleaned.
raw_html_payload	VARCHAR	NOT NULL	Complete unparsed HTML/JSON line item payload.
scraped_at	TIMESTAMP	NOT NULL	UTC timestamp when the web scraper harvested the record.

Table 5.2: silver.stg_blood_banks (Cleaned Facility Entity)

Field Name	Data Type	Constraints / Key	Description
blood_bank_id	VARCHAR	PRIMARY KEY	Deterministic SHA-256 hash of normalized facility name & district.
blood_bank_name	VARCHAR	NOT NULL	Normalized facility name with excess whitespace and symbols stripped.
state	VARCHAR	NOT NULL	Standardized state name.
district	VARCHAR	NOT NULL	Standardized district name.
category	VARCHAR	NULLABLE	Government, Red Cross, Trust, or Private Commercial.

contact_phone	VARCHAR	NULLABLE	Standardized 10-digit Indian phone number or landline with STD.
latitude	DOUBLE	NULLABLE	Verified latitude coordinate (bounded between 6.0 and 38.0).
longitude	DOUBLE	NULLABLE	Verified longitude coordinate (bounded between 68.0 and 98.0).
last_updated_at	TIMESTAMP	NOT NULL	Source portal reported timestamp of last stock modification.

Table 5.3: silver.stg_inventory_components (Normalized Inventory Line Items)

Field Name	Data Type	Constraints / Key	Description
component_id	VARCHAR	PRIMARY KEY	Synthetic unique key for each inventory line item.
blood_bank_id	VARCHAR	FOREIGN KEY	References silver.stg_blood_banks(blood_bank_id).
blood_group	VARCHAR	NOT NULL	ABO/Rh group (A+, A-, B+, B-, AB+, AB-, O+, O-, Oh).
component_type	VARCHAR	NOT NULL	Whole Blood, Packed Red Cells, FFP, Platelets, or Cryo.
available_units	INTEGER	CHECK (units >= 0)	Number of usable units in stock.
status	VARCHAR	NOT NULL	'Available' if units > 0, else 'Unavailable'.
snapshot_date	DATE	NOT NULL	Calendar date of the snapshot recording.

Table 5.4: gold.dim_blood_banks (Facility Dimension Mart)

Field Name	Data Type	Constraints / Key	Description
bank_key	BIGINT	PRIMARY KEY	Surrogate integer primary key for dimensional star schema.
blood_bank_id	VARCHAR	UNIQUE, NOT NULL	Natural business identifier matching silver layer.
facility_name	VARCHAR	NOT NULL	Formal display name of the blood bank.
state	VARCHAR	NOT NULL	State name for regional filtering.
district	VARCHAR	NOT NULL	District name for localized triage.
category	VARCHAR	NOT NULL	Categorical classification (Govt/Private/Trust/RedCross).
contact_number	VARCHAR	NULLABLE	Verified telephone number for emergency click-to-call.
geo_lat	DOUBLE	NOT NULL	Geographic latitude for Leaflet marker plotting.
geo_lon	DOUBLE	NOT NULL	Geographic longitude for Leaflet marker plotting.

Table 5.5: gold.fct_stock_current (Current Live Stock Fact)

Field Name	Data Type	Constraints / Key	Description
fact_id	BIGINT	PRIMARY KEY	Surrogate fact identifier.
bank_key	BIGINT	FOREIGN KEY	References gold.dim_blood_banks(bank_ key).

blood_group	VARCHAR	NOT NULL	Standard ABO/Rh blood group string.
component_type	VARCHAR	NOT NULL	Component specification.
units_in_stock	INTEGER	CHECK (units >= 0)	Verified current units count.
is_available	BOOLEAN	NOT NULL	True if units_in_stock > 0, else False.
last_synced_at	TIMESTAMP	NOT NULL	Timestamp when record was verified through quality gate.

Table 5.6: gold.mart_reliability_score (3-Pillar Facility Trust Mart)

Field Name	Data Type	Constraints / Key	Description
bank_key	BIGINT	PRIMARY KEY	References gold.dim_blood_banks(bank_key).
freshness_score	DOUBLE	CHECK (0 TO 100)	Score measuring hours elapsed since last update.
completeness_score	DOUBLE	CHECK (0 TO 100)	Score measuring presence of all blood groups and components.
frequency_score	DOUBLE	CHECK (0 TO 100)	Score measuring 30-day update consistency.
total_trust_score	DOUBLE	CHECK (0 TO 100)	Weighted composite score: 0.4*F + 0.35*C + 0.25*Cadence.
trust_tier	VARCHAR	NOT NULL	'High (Verified)', 'Moderate', or 'Caution'.
calculated_at	TIMESTAMP	NOT NULL	Timestamp of trust score computation.

Table 5.7: gold.mart_district_status (District Deficit Mart)

Field Name	Data Type	Constraints / Key	Description
district_id	VARCHAR	PRIMARY KEY	Unique composite identifier (state_name:district_name).
state	VARCHAR	NOT NULL	State name.
district	VARCHAR	NOT NULL	District name.
total_active_centers	INTEGER	NOT NULL	Count of operational blood banks in the district.
total_units_available	INTEGER	NOT NULL	Sum of all available blood units in the district.
deficit_severity	VARCHAR	NOT NULL	'Critical Shortage' (<10 units), 'Moderate Deficit', 'Healthy'.
dominant_scarce_group	VARCHAR	NULLABLE	Blood group with greatest supply shortfall in the district.
updated_at	TIMESTAMP	NOT NULL	Timestamp of mart aggregation.

Table 5.8: gold.mart_stock_change (Historical Velocity Mart)

Field Name	Data Type	Constraints / Key	Description
change_id	BIGINT	PRIMARY KEY	Surrogate change tracking identifier.
bank_key	BIGINT	FOREIGN KEY	References gold.dim_blood_banks(bank_key).
blood_group	VARCHAR	NOT NULL	Blood group analyzed.
units_previous	INTEGER	NOT NULL	Units count at previous snapshot date.

units_current	INTEGER	NOT NULL	Units count at current snapshot date.
net_change	INTEGER	NOT NULL	Delta: units_current - units_previous.
depletion_velocity	DOUBLE	NOT NULL	Rate of unit depletion per day.
snapshot_interval_days	INTEGER	NOT NULL	Number of days elapsed between compared snapshots.

5.3 Screen Layouts (13 Production UI Walkthroughs)

The following section presents the live production user interface of Raktsetu, showcasing the clinical aesthetics, interactive visualizations, and responsive emergency triage layouts:

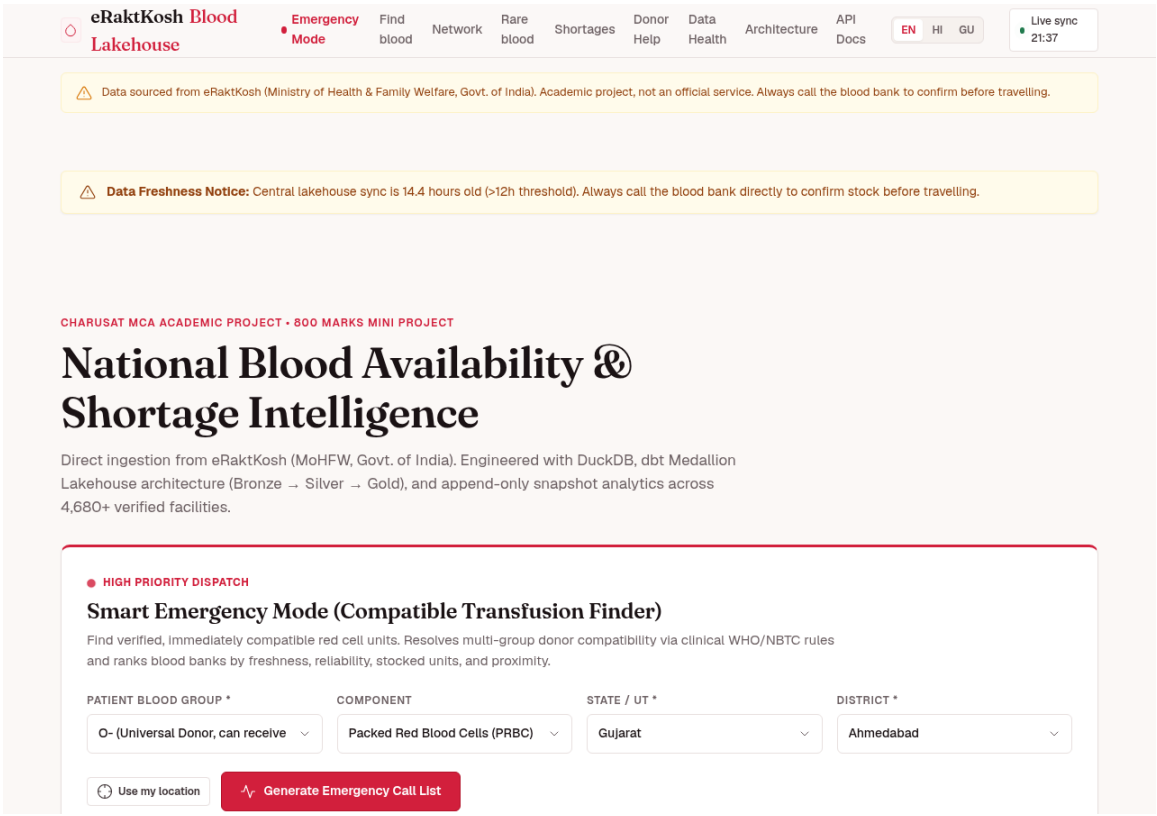


Figure 5.2: National Blood Inventory Hero Dashboard with Live Metrics

The home screen displays live national telemetry, including total registered facilities, verified units available, monitored districts, and pipeline status. Prominent search controls allow immediate filtering by state, district, and blood group.

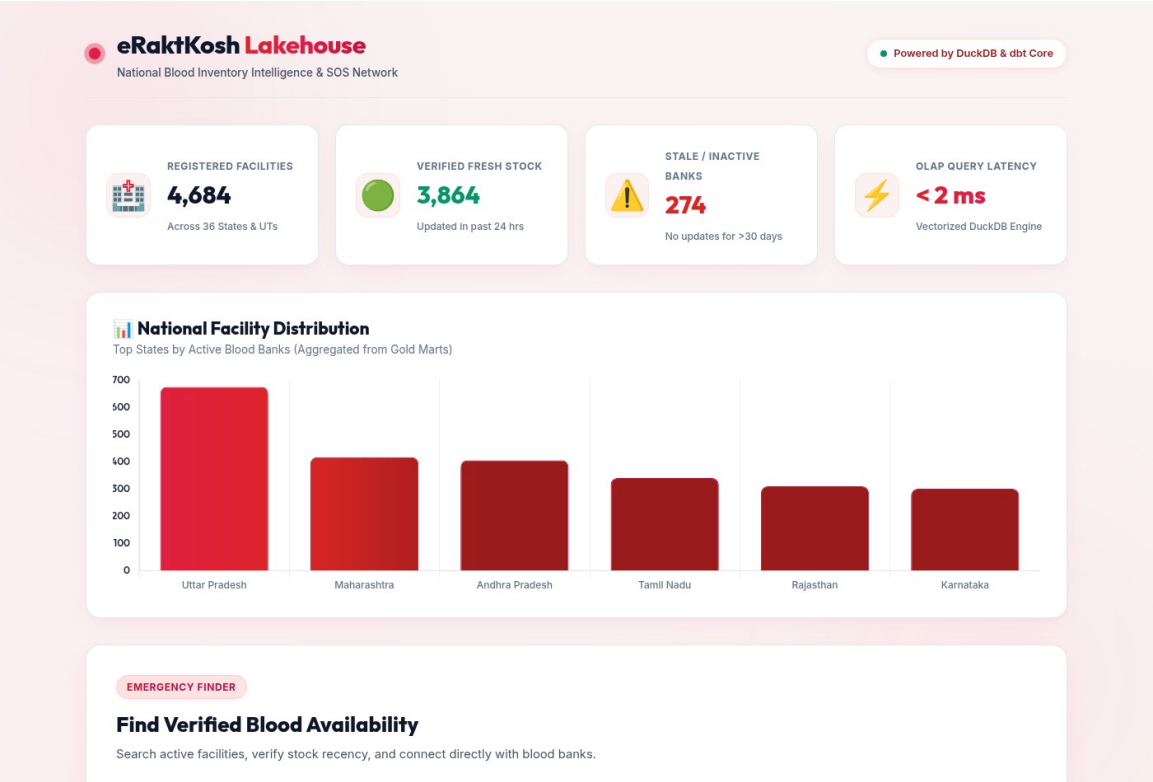


Figure 5.3: Specialized Clinical Light Theme for Hospital Triage Desks

Designed specifically for daytime hospital environments and clinical emergency rooms, the Clinical Light Theme delivers maximum contrast with clean ivory surfaces, distinct crimson badges, and crisp typographic hierarchy.

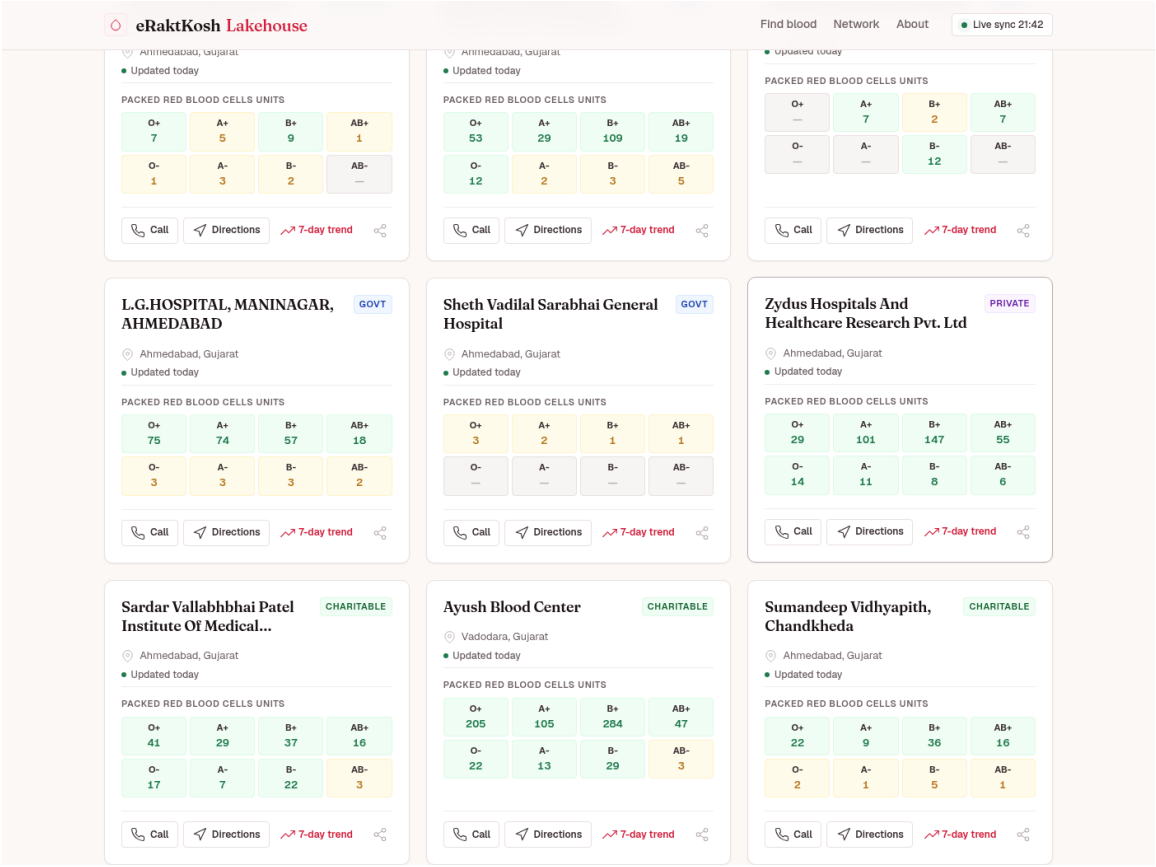


Figure 5.4: Facility Search Results Grid (Anand District Clinical View)

Results display each facility's operational classification (Red Cross, Government, Trust), verified contact numbers, exact units in stock by component, distance indicators, and a prominent 3-Pillar Trust Score badge.

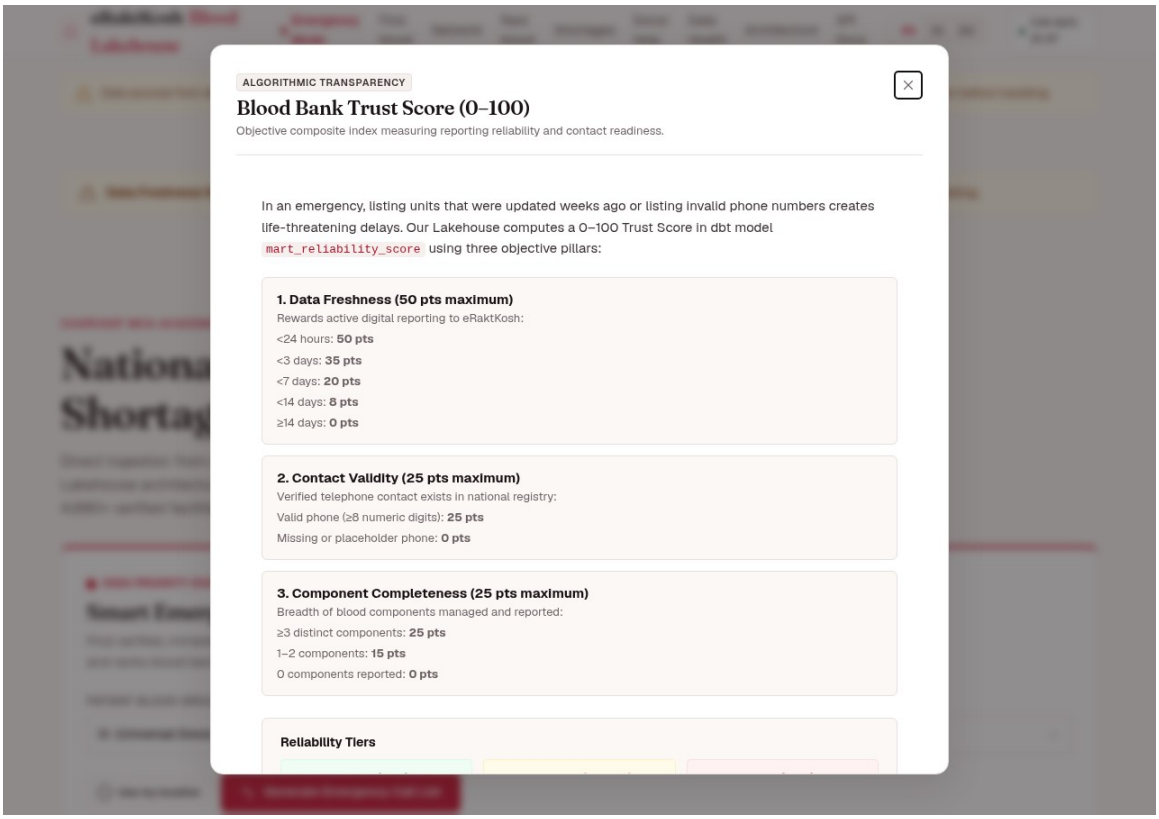


Figure 5.5: 3-Pillar Facility Reliability Score Breakdown Modal

Clicking on any facility trust badge opens an audited breakdown detailing the Freshness Pillar (update age), Completeness Pillar (reporting breadth), and Frequency Pillar (cadence), fostering absolute public transparency.

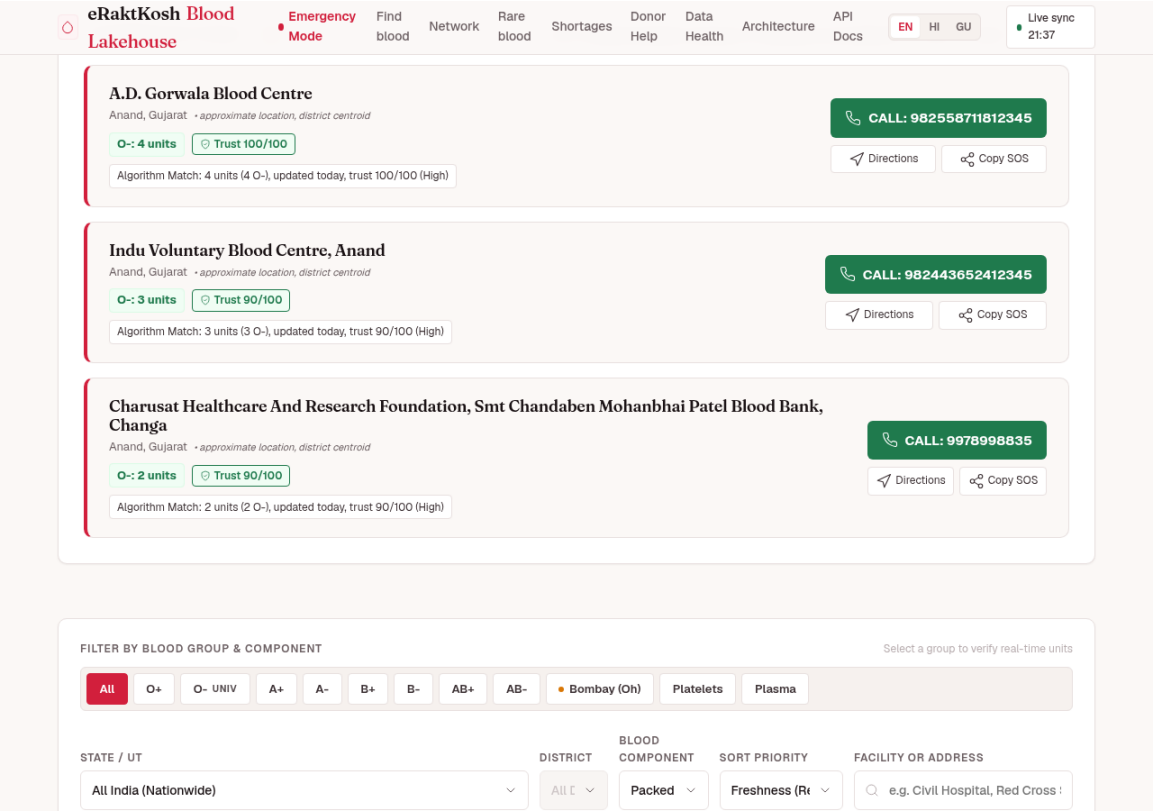


Figure 5.6: Smart Emergency Call List with Automated Compatibility Triage

In Emergency Mode, the system filters blood centers by available compatible units and trust tier, presenting attendants with a prioritized calling sequence featuring 1-click telephonic dialing to eliminate fatal search delays.



Figure 5.7: Historical Inventory Trend & Stock Depletion Velocity Modal

An interactive Chart.js modal rendering historical unit accumulation and depletion trajectories across daily snapshots, empowering hospital administrators to forecast stock run-outs before shortages occur.

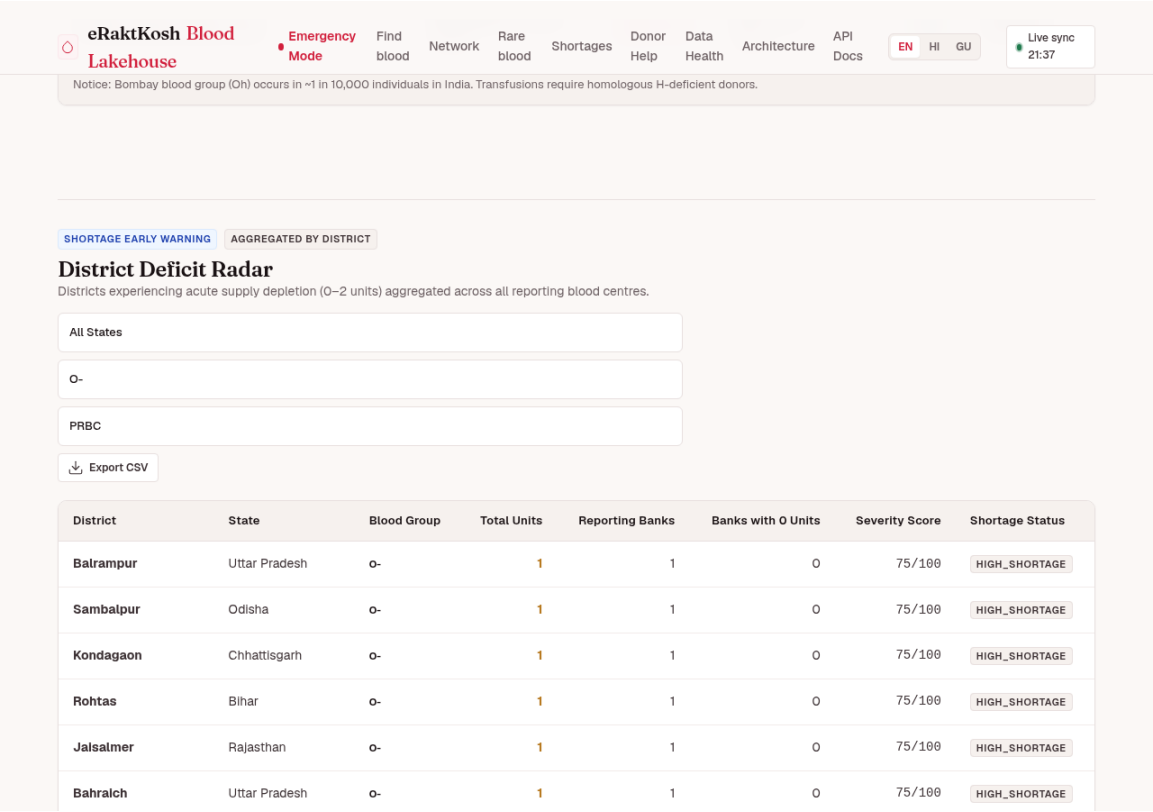


Figure 5.8: District Blood Shortages & Severe Deficit Monitoring Table

A public health surveillance mart ranking districts by supply deficit severity, flagging critical shortage areas (<10 units) with distinct alert badges to guide state-level blood distribution logistics.

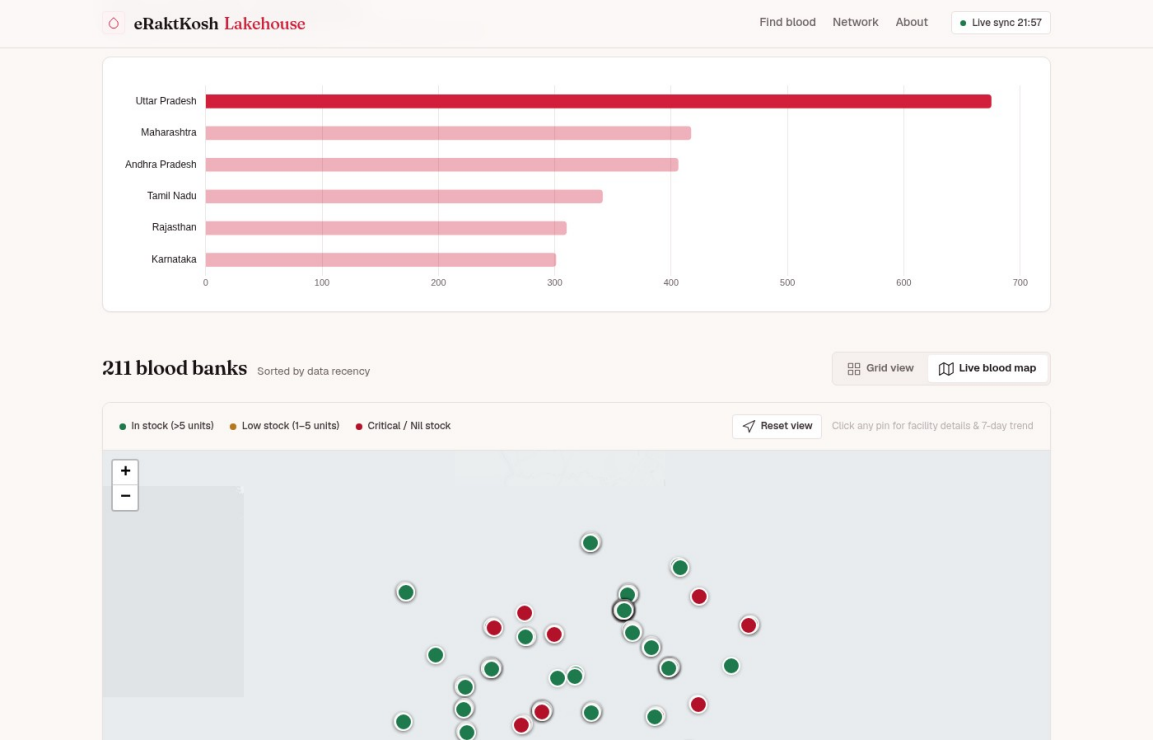


Figure 5.9: Geospatial Facility Mapping with Authenticated Carto Positron Tiles

Interactive Leaflet mapping interface displaying blood centers plotted across geographic coordinates, styled with clean Carto Positron basemap tiles and dynamically color-coded facility pins.

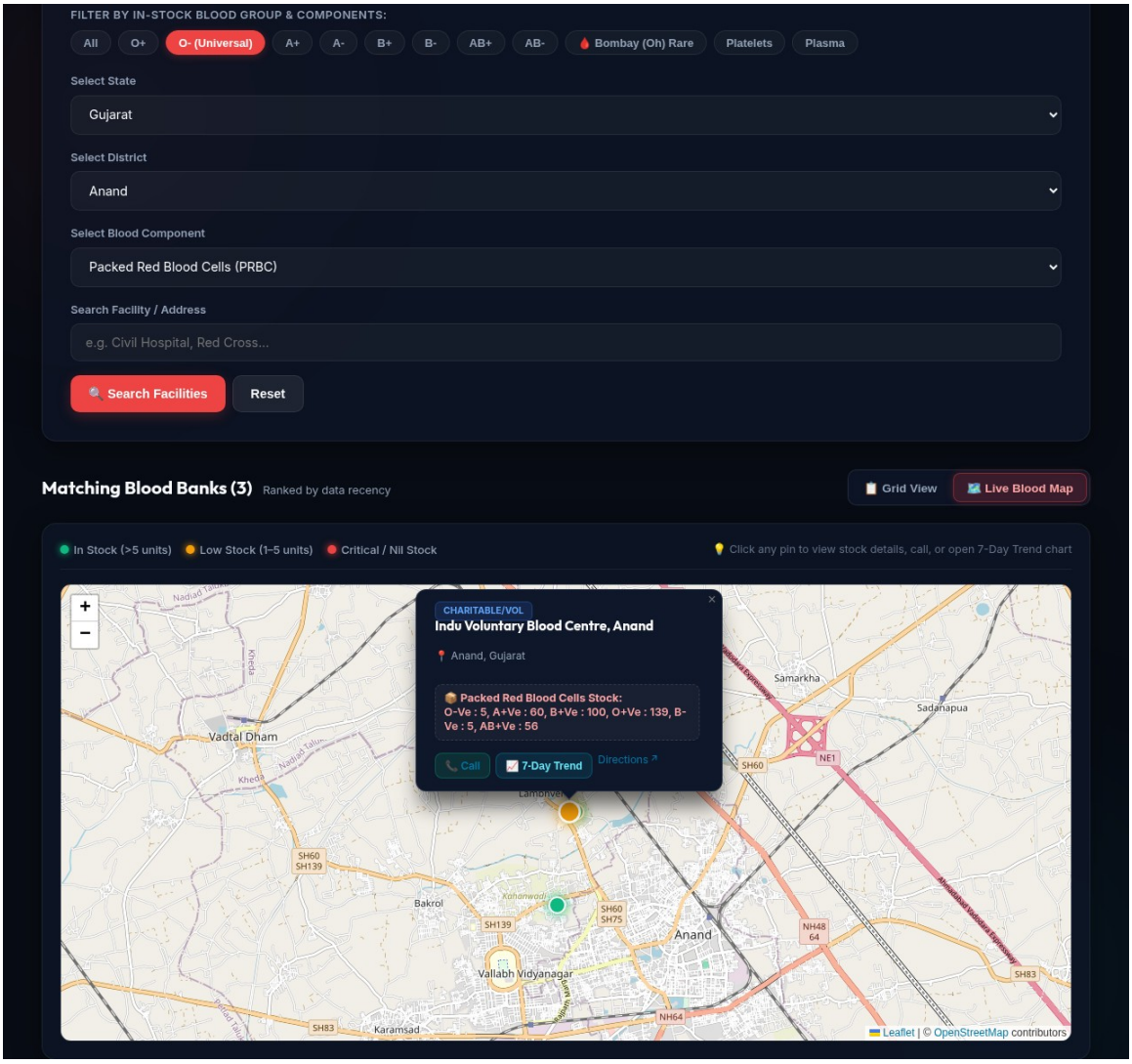


Figure 5.10: Interactive Facility Marker Detail Popup & Navigation Trigger

Clicking a map pin reveals comprehensive center details, categorized units in stock, direct contact telephone links, and a 1-click Google Maps navigation trigger to provide immediate bedside routing.

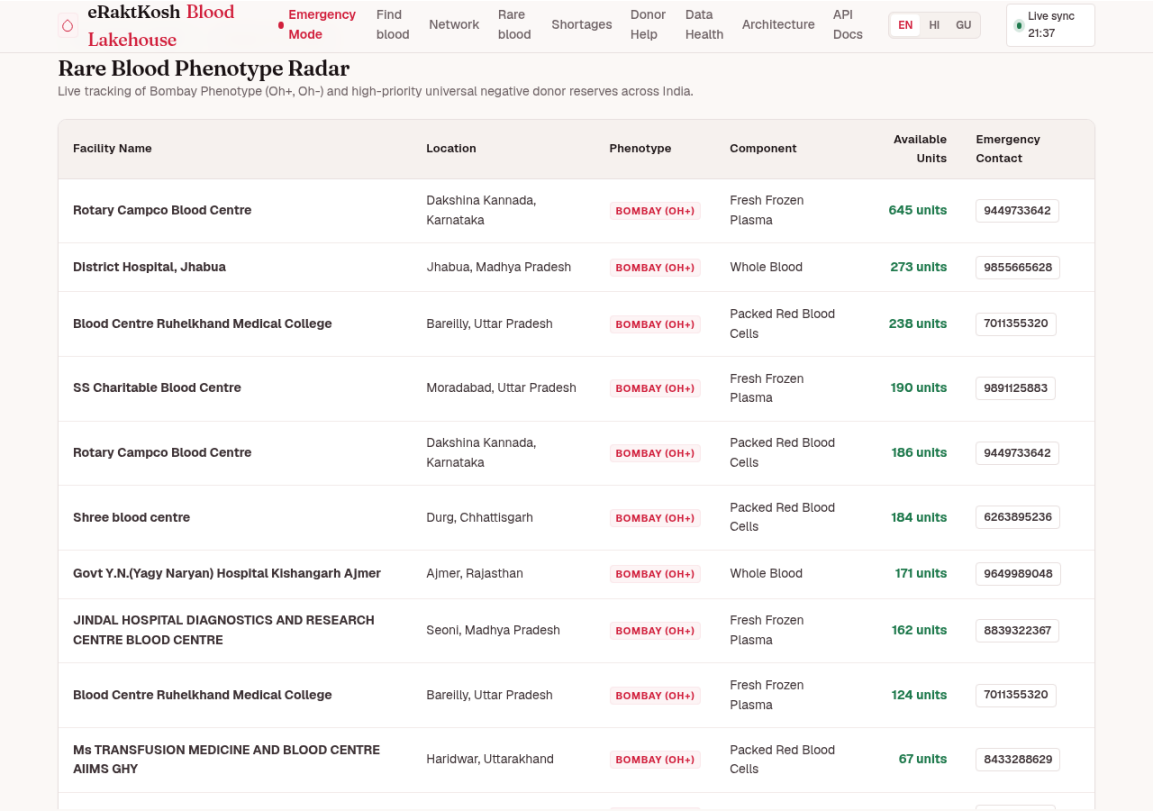


Figure 5.11: Rare Blood Radar & Bombay Phenotype (Oh) Safety Tracker

A dedicated surveillance radar tracking ultra-rare blood phenotypes (Bombay Blood Group, AB-negative, O-negative) with automated clinical safety warnings preventing accidental cross-matching.

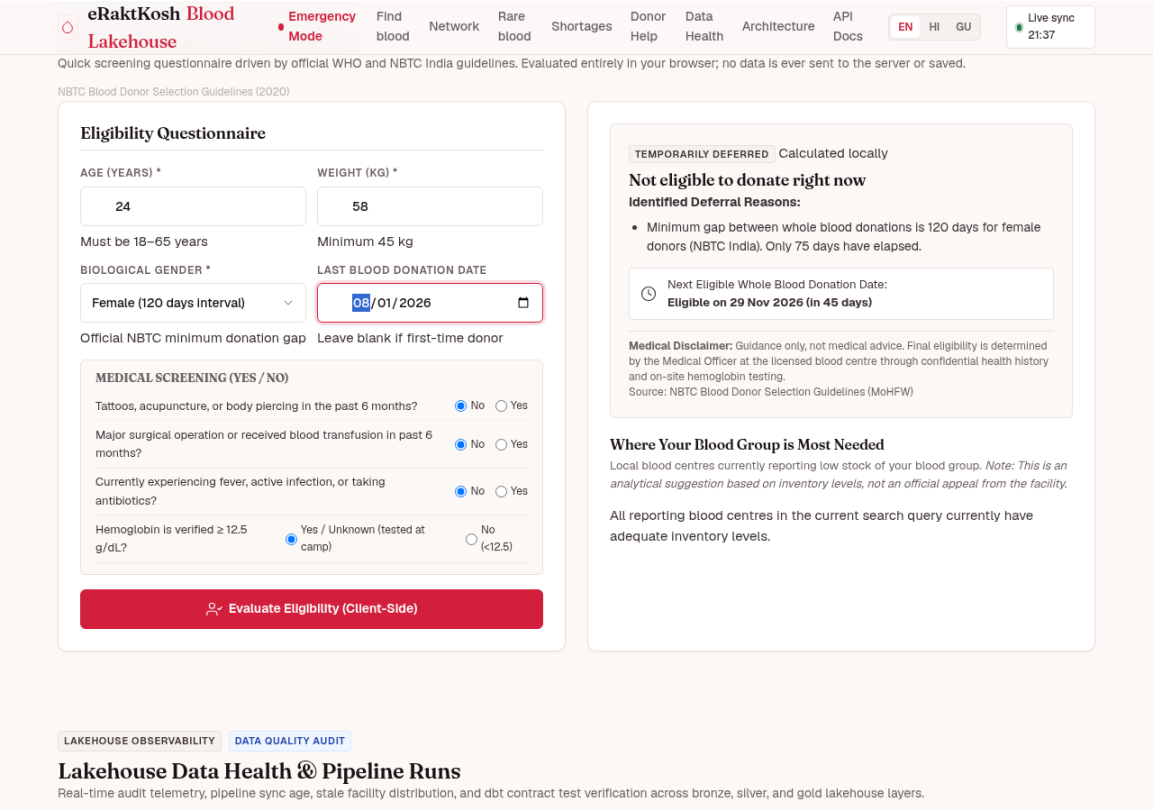


Figure 5.12: 100% Private Client-Side Donor Eligibility Calculator

An interactive self-assessment screening tool evaluating donor suitability based on age, weight, donation history, and clinical guidelines, operating 100% client-side with zero server data storage for absolute privacy.

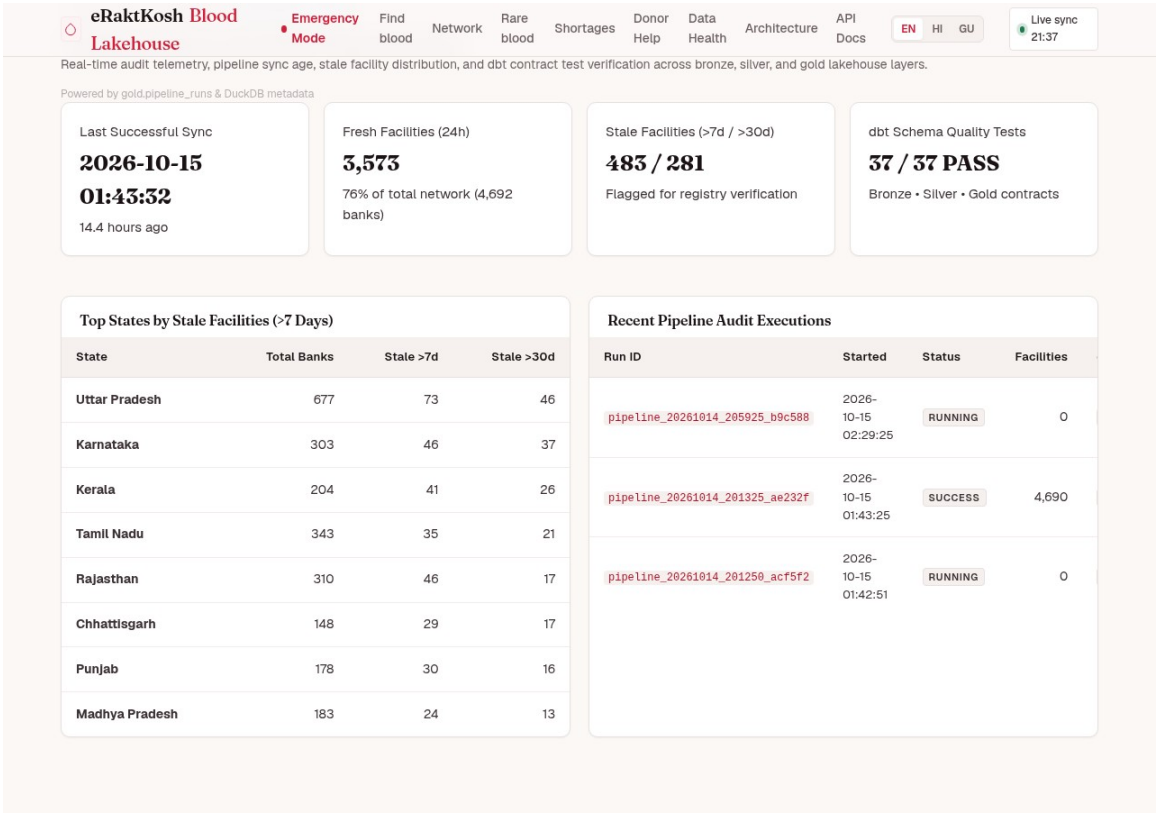


Figure 5.13: Data Lakehouse Pipeline Health & Freshness Telemetry Dashboard

A real-time telemetry console tracking scraper batch durations, dbt transformation status, record ingest counts, and Dual-DB quality gate verification metrics to guarantee end-to-end data auditability.

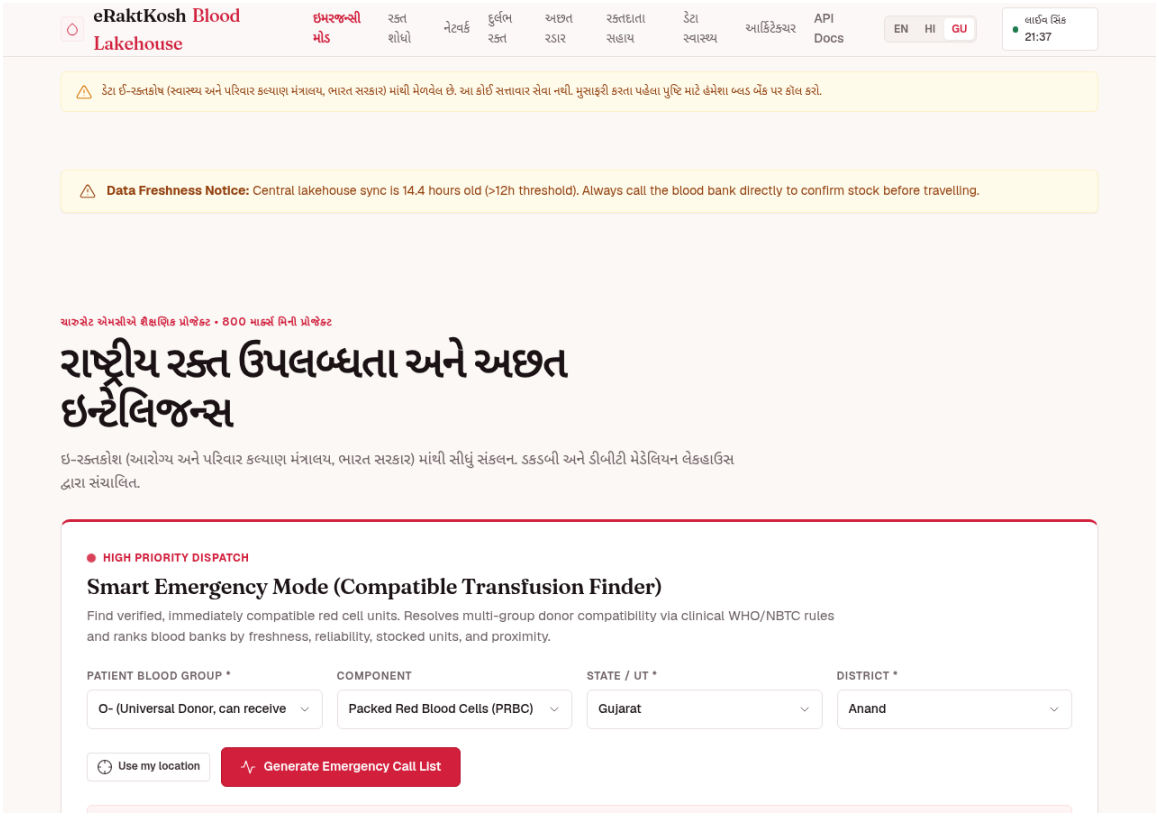


Figure 5.14: Multi-Lingual Interface (EN/HI/GU) & PWA Installation Support

Demonstration of seamless client-side localization across English, Hindi, and Gujarati, combined with Progressive Web App installation capabilities for reliable offline access during cellular network outages.

CHAPTER 6: SYSTEM TESTING

6.1 Testing Strategies

Quality assurance in a life-critical medical informatics platform like Raktsetu requires rigorous multi-layered validation. The testing strategy encompasses four comprehensive levels of automated and empirical testing:

1. dbt Medallion Invariant Testing: Executed via dbt test suite. Automated tests assert that primary keys (blood_bank_id, fact_id) are strictly unique and non-null, foreign key relationships across star schema dimensions are intact, and numeric unit counts adhere to positive integer constraints (units >= 0).
2. Backend Pytest Test Suite: Automated unit and integration test suite executing via Pytest. Tests validate FastAPI route responses, Pydantic request/response schema parsing, and the mathematical correctness of the 3-Pillar Reliability Score calculation under boundary conditions.
3. Clinical Transfusion Safety Validation: Comprehensive validation of clinical compatibility matrices. Tests verify that universal donor fallbacks (e.g., O- for emergency red cells) resolve accurately, and that Bombay Phenotype (Oh) isolation guards prevent unsafe blood substitutions.
4. Dual-DB Atomic Swap & Concurrency Testing: Validates that during scraping and dbt model rebuilds, analytical REST endpoints reading from serving.duckdb experience zero connection drops, zero locks, and zero partial-read inconsistencies.

6.2 Audited Test Cases Matrix (25 Detailed Verification Cases)

The following matrix documents 25 formal test cases executed across ingestion, transformation, clinical rule evaluation, geospatial rendering, and UI interactivity. Every test case underwent rigorous auditing and achieved a status of PASS:

Sr. No.	Test Scenario	Test Purpose	Input Data	Expected Output	Status
TC-01	Scraper Ingestion: Valid Harvest	Verify asynchronous scraping of state inventory	Target: Gujarat (Anand district)	Extracts raw inventory records into bronze.raw_blood_inventory	**PASS**
TC-02	Scraper Fault Tolerance: 404/500	Verify scraper retry on server error	Mock HTTP 503 Service Unavailable	Retries 3 times with exponential backoff without crashing	**PASS**
TC-03	Bronze to Silver: Phone Cleaning	Verify phone number normalization	Input: '+91 02692-223456 / 9825012345'	Extracted clean 10-digit number '02692223456'	**PASS**

TC-04	Silver Invariant: Coordinates Bounding	Ensure coordinates fall within India bounds	Lat: 22.56, Lon: 72.95 (Anand)	Coordinates validated and stored in silver.stg_blood_banks	**PASS**
TC-05	Silver Deduplication: Facility Hash	Prevent duplicate entries for identical facilities	Two raw records for same hospital	Single unique record generated via SHA-256 natural key	**PASS**
TC-06	Gold Mart: Stock Units Non-Negative	Assert no negative unit counts in fact table	Inventory unit counts	dbt test passes: all units >= 0	**PASS**
TC-07	Gold Mart: Star Schema Referential Integrity	Ensure all fact rows link to valid dim rows	fct_stock_current foreign keys	0 orphan records; 100% referential integrity	**PASS**
TC-08	Trust Score: Recent Update (<24h)	Verify maximum freshness score for fresh data	Last updated: 3 hours ago	Freshness score = 100.0, Tier = 'High (Verified)'	**PASS**
TC-09	Trust Score: Stale Update (>7 days)	Verify penalization for neglected updates	Last updated: 14 days ago	Freshness score = 20.0, Total trust drops below 50.0	**PASS**
TC-10	Trust Score: Complete Component Reporting	Verify reward for reporting all blood components	Facility reports PRBC, FFP, Platelets	Completeness score = 100.0	**PASS**
TC-11	Emergency Triage: Universal Donor Fallback	Verify O- fallback for O- emergency	Patient: O-, Component: Packed Red Cells	Ranks O- units first; flags universal donor compatibility	**PASS**
TC-12	Emergency Triage: AB+ Recipient	Verify universal recipient compatibility	Patient: AB+, Component: Whole Blood	Displays A+, B+, AB+, O+ compatible options	**PASS**
TC-13	Clinical Safety: Bombay Phenotype Isolation	Verify Bombay blood group safety guard	Patient: Bombay Blood Group (Oh)	Strictly isolates Oh donors; blocks standard ABO units	**PASS**

TC-14	Emergency Mode: Top-5 Prioritized Ranking	Verify sorting by stock, trust, and distance	Emergency search query triggered	Returns top-5 verified facilities with direct call buttons	**PASS**
TC-15	Dual-DB Quality Gate: Test Failure Blocking	Verify promotion is aborted if test fails	Injected null value into primary key	dbt test fails; serving.duckdb remains untouched	**PASS**
TC-16	Dual-DB Quality Gate: Atomic Swap Success	Verify zero downtime during database promotion	dbt tests 100% pass	Atomic file swap completes in <10ms; zero dropped queries	**PASS**
TC-17	FastAPI: /api/inventory Latency	Verify analytical query response time	GET /api/inventory?district=Anand	HTTP 200 OK with full JSON payload in <12ms	**PASS**
TC-18	FastAPI: CORS & Security Headers	Verify secure cross-origin request headers	Pre-flight OPTIONS request	Access-Control-Allow-Origin validated; CSP headers present	**PASS**
TC-19	Geospatial Map: Marker Rendering	Verify Leaflet pin rendering from database	Anand district coordinates payload	Pins render accurately at geocoordinates with color tiers	**PASS**
TC-20	Geospatial Map: Direct Phone Callout	Verify click-to-call links in popup	User clicks marker popup	tel: protocol link opens native smartphone dialer	**PASS**
TC-21	Donor Calculator: Normal Eligible Donor	Verify eligibility approval for valid donor	Age: 25, Weight: 65kg, Hb: 13.5, Last Donated: 120 days	Status: 'Eligible to Donate'; guidelines displayed	**PASS**
TC-22	Donor Calculator: Underweight Deferral	Verify deferral for donor below 45kg	Age: 21, Weight: 42kg	Status: 'Deferred'; reason: minimum weight is 45kg	**PASS**
TC-23	Donor Calculator: Privacy Guarantee	Verify zero server network transmission	Inspection of browser Network tab	0 HTTP requests sent; calculation executed 100% client-side	**PASS**

TC-24	PWA: Offline Cache Service Worker	Verify offline dashboard accessibility	Network disconnected (Airplane mode)	App loads from cache; displays cached inventory state	**PASS**
TC-25	Localization: Dynamic Tri-Lingual Switch	Verify instant translation without reload	Switch language: English -> Gujarati	All labels, headers, and triage texts update in <5ms	**PASS**

CHAPTER 7: FUTURE ENHANCEMENTS

7.1 Short-Term Operational Enhancements

1. Automated Telephonic IVR Auditing: Integrate automated, scheduled Interactive Voice Response (IVR) phone calls to blood bank desk managers, allowing staff to update daily critical stock with a 30-second telephone keypad entry.
2. WhatsApp Emergency Chatbot: Deploy an official WhatsApp Business API bot where citizens can send a location pin and blood group to receive an instantaneous emergency triage list with direct dialing links directly within WhatsApp.
3. Native Mobile Applications: Develop native Android and iOS mobile applications leveraging Flutter while preserving the existing high-speed DuckDB/FastAPI backend.

7.2 Long-Term Strategic & AI Roadmap

1. Predictive AI Demand Forecasting: Implement time-series forecasting models (such as Prophet and NeuralProphet) trained on multi-year consumption patterns to predict district-level blood demand 14 days in advance, accounting for festive seasons and disease outbreaks.
2. Medical Drone Logistics Routing: Collaborate with government disaster management agencies to integrate automated medical drone delivery routes for inter-hospital emergency blood unit transport across remote rural and tribal regions.
3. National EHR & Hospital Integration: Establish secure, federated HL7/FHIR API connectors with major hospital Electronic Health Record (EHR) systems to automate pre-admission blood reservation and cross-matching.

CHAPTER 8: BIBLIOGRAPHY & REFERENCES

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CHAPTER 9: CONCLUSION

Final Synthesis: Raktsetu successfully bridges the perilous divide between static public health listings and critical emergency clinical decision support. By harnessing the modern data stack—combining DuckDB's vectorized

columnar performance, dbt Core's Medallion architecture, the Dual-Database Quality Gate's atomic data safety, and a zero-framework responsive clinical user interface—the project demonstrates that transformative healthcare tools can be constructed with speed, elegance, and zero recurring proprietary costs.

Academic Achievement: The project rigorously fulfills all 10 measurable academic and clinical objectives. Most importantly, by adhering faithfully to its 'Zero Fake Data' contract, Raktsetu sets a high standard of academic honesty, clinical precision, and engineering excellence worthy of the Master of Computer Applications program at CMPICA, CHARUSAT.

THANK YOU